# Learning Curves

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

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TEACHER GUIDE · 1 OF 9

## Learning Curves

> Prepare Learning Curves: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Read **training**-size **learning curve**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

**Repeat**ed copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Overfitting

Without opening the old lesson, explain one key idea from Over**fitting**. Give an example and a mistake that the idea helps you avoid.

Look for: Compare **training** and validation **evidence** for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

### Metrics Match Consequences

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a declared score threshold including equality

Reference example, not the only acceptable wording: Teacher explanation: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

### Train a Regression Model

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a line from a declared **training** candidate set

Reference example, not the only acceptable wording: Teacher explanation: A 5 B 3 C 0; choose m 2 b 1.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

**Training** sizes 20,40,80 yield validation accuracies 62%,**70%**,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

### Concept to explain

**Learning curve**s plot performance against **training** size or effort and help distinguish under**fitting**, over**fitting**, and data limitation.

| Term | Meaning |
| --- | --- |
| learning curve | performance across training sizes |
| bias | systematic underfitting |
| variance | sensitivity to particular samples |
| plateau | little improvement from added data |

### Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

## Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on **learning curve**s and **variability**. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

### Readiness answer

**Evidence** about examples that did not fit the model parameters.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

### End goal for the pictorial case

By the end, I can explain **learning curve**s using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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## Explain the mechanism

> Model Learning Curves using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

### Question to explain

QUESTION Interpret three curve patterns and select a targeted next experiment.

MODEL IDEA **Learning curve**s plot performance against **training** size or effort and help distinguish under**fitting**, over**fitting**, and data limitation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

### How the mechanism works

UNDER THE HOOD Both train and validation scores are low and close: likely under**fitting**. Train high, validation much lower: likely high **variance** or over**fitting**.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

### Step 1

STEP 1 Both train and validation scores are low and close: likely under**fitting**.

### Step 2

STEP 2 Train high, validation much lower: likely high **variance** or over**fitting**.

### Step 3

STEP 3 Validation improves as data grows and gap narrows: more representative data may help.

### Step 4

STEP 4 Curves suggest hypotheses; confirm with controlled experiments and error slices.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

**Concrete teaching case**

### The teaching case

With20,40,80 **training** leaves, fixed-set validation accuracy is**60%**,**70%**,78%.

### The teaching case

Interpret the curve and uncertainty.

**Exact explanation to model**

### Why the result follows

These runs improve with more data. Further representative data may help, but gains are not guaranteed. Controlled **repeat**s reveal **variability**.

**Explain the added worked example**

## Explain the added worked example · 1

A **learning curve** asks how performance changes as the amount of **training evidence** grows. Keep the validation protocol, model settings and metric comparable.

Use subsets of **training** only; never add validation rows into the **training** subsets. Multiple **repeat**s reveal sensitivity to sample selection or model randomness.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **learning curve**s and **variability**.

## Explain the added worked example · 2

At **training** sizes 10, 20, 40, validation MAEs across three declared **repeat**s are [7, 9, 8], [5, 7, 6], [4, 6, 5]. Means are 8, 6, 5 and **range**s 2, 2, 2. Mean **training** MAEs are 1, 2, 3.

Validation improves as **training** grows, while **training** error rises because **fitting** more varied examples can be harder.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

### Key terms

| Term | Meaning |
| --- | --- |
| Learning curve | Performance plotted against training sample size |
| Repeat | Another run under a recorded procedure |
| Variability | How much results differ between runs |
| Range | Maximum minus minimum of observed values |

## Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

### Learning curves and variability at a glance

| PART | EVIDENCE OR ACTION | INTERPRETATION |
| --- | --- | --- |
| 10 training rows | Validation 7 9 8 | Mean 8 range 2 |
| 20 training rows | Validation 5 7 6 | Mean 6 range 2 |
| 40 training rows | Validation 4 6 5 | Mean 5 range 2 |

The decreasing gap is encouraging in this experiment. It does not establish that doubling again will halve error. A range from three repeats is a descriptive spread, not a confidence interval or a guarantee that future results fall inside it. Repeated runs can share records and therefore are not independent studies.

**Why can training error rise while validation error falls? Explain the steps or evidence behind your response.**

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Learning Curves

Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

### The facts we will use

#### Supplied facts

At 10 **training** plants validation errors across3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set.

#### Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/28

# Learning Curves

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Summarise first size

Mean6 errors; range5-7, corresponding to mean 40% accuracy.





## 2. Summarise larger size

Mean4 errors; range3-5, corresponding to mean 60% accuracy.





## 3. State bounded pattern

More training plants improved the average in these supplied runs; variability remains.

**CHANGE ONE THING**

A30-plant experiment gives errors [4, 4, 4].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Summarise first size | Mean6 errors; range5–7, corresponding to mean 40% accuracy. |
| 2. Summarise larger size | Mean4 errors; range3–5, corresponding to mean 60% accuracy. |
| 3. State bounded pattern | More training plants improved the average in these supplied runs; variability remains. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### Why the result follows

Read the picture in order. More **training** plants improved the average in these supplied runs; **variability** remains. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

#### Keep the stated limit

These toy runs do not establish that data quantity alone caused improvement; collection quality and conditions matter.

### Change one thing in the pictured case

#### Change one thing

Change: A30-plant experiment gives errors [4, 4, 4].

#### Predicted result

**Prediction**: Mean remains4; this additional size does not improve the observed mean over20 plants.

#### Why the result follows

Reason: **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

### Ask while showing the picture

## Ask while showing the picture · 3

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

# Demonstrate and guide practice

> Demonstrate Learning Curves, reduce support deliberately and check individual reasoning before discussing answers.

**Board sequence**

### 1. Put the facts on the board

Write the given facts: With20,40,80 **training** leaves, fixed-set validation accuracy is**60%**,**70%**,78%.

### 2. State the decision to explain

Write the question to explain: Interpret the curve and uncertainty.

### 3. Reveal the worked reasoning

Reveal after **prediction**s: These runs improve with more data. Further representative data may help, but gains are not guaranteed. Controlled **repeat**s reveal **variability**.

### Hold back the independent task

Keep for an independent attempt: As **training** examples grow from 50 to 100 to 200, validation accuracy is **60%**, 72%, 78%. What does the curve suggest, and what does it not guarantee?

### Classroom dialogue

#### Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

#### Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### Model these guided questions

Guided response: More varied **training** cases may be harder to fit exactly while giving a pattern that generalizes better.

Exit response: It only describes the smallest and largest observed **repeat**s; no coverage-probability calculation or independent-sampling argument was supplied.

### Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: What does validation loss add to training loss? Check the explanation before proceeding.

> **Stage:** 7 to 18 minutes

> **Teacher action:** Read the opening and model the worked evidence. Ask students to identify learning curve in context.

> **Stage:** 18 to 32 minutes
>
> **Teacher action:** Complete W 1 and W 2 with support: summarize repeats; read both curves. Require the evidence, not only the final response.

> **Stage:** 32 to 45 minutes
>
> **Teacher action:** Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

> **Stage:** Session 2 start to 8 minutes
>
> **Teacher action:** Retrieve the distinction between learning curve and repeat. Invite a counterexample to an overconfident claim.

> **Stage:** 8 to 20 minutes
>
> **Teacher action:** Discuss the application and complete W 3: Can the 40-row result guarantee a mean MAE of 2.5 at 80 rows?

> **Stage:** 20 to 35 minutes
>
> **Teacher action:** Collect W4 independently before giving hints. Assess these outcomes: Read training-size learning curves; Describe variability across repeated development runs. Record any support used.

> **Stage:** 35 to 45 minutes
>
> **Teacher action:** Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

### Reduce help deliberately

#### Completion practice

Completion task: Why can **training** error rise while validation error falls?

#### Reasoning to listen for

Teacher check: Teacher explanation: More varied **training** cases may be harder to fit exactly while giving a pattern that generalizes better.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for **Learning curve**s and **variability**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

#### Expected repair

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Mean(5+7+6)/3=6; **range** 7-5=2.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### Pictorial practice question

What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

### Reasoned teacher answer

The raw change is confounded by changed evaluation cases. Prefer a fixed appropriately reserved validation set and record **repeat**ed-run variation.

Changed-case check: Mean remains4; this additional size does not improve the observed mean over20 plants. **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

Changed-case reasoning: Mean remains4; this additional size does not improve the observed mean over20 plants. **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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## Run the weekly evidence lab

> Run the Learning Curves investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Curve Clinic: match three graphs to diagnoses, challenge each diagnosis, and propose the smallest informative experiment.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

### Diagnose and reteach

MISCONCEPTION TO REPAIR Every **learning curve** that rises proves more data will solve the problem.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Mean(5+7+6)/3=6; **range** 7-5=2.

A stronger claim makes a stronger project: It only describes the smallest and largest observed **repeat**s; no coverage-probability calculation or independent-sampling argument was supplied.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: Means 7 and 6; **range**s 2 and 3.

The larger condition improves the mean by 1 but has greater observed spread in these three **repeat**s.

### Match the support to the first error

| What the response shows | Next teaching action |
| --- | --- |
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Learning Curves questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Worked case answer

#### Teacher key · Worked case answer

These runs improve with more data. Further representative data may help, but gains are not guaranteed. Controlled **repeat**s reveal **variability**.

### Independent question

#### Independent question · withhold the key

As **training** examples grow from 50 to 100 to 200, validation accuracy is **60%**, 72%, 78%. What does the curve suggest, and what does it not guarantee?

### Expected independent answer

#### Teacher key · Expected independent answer

More representative data helped over these measured sizes, with smaller later gains. It does

not guarantee continued improvement or performance in untested conditions.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “An improving **learning curve** guarantees unlimited future gains.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

#### Teacher key · Original transfer answer

No. Gains need not be linear or continue. Keep validation fixed, add representative **training** examples, record exact subsets and **repeat** stochastic runs.

Compare variation and collection costs; do not imply the line predicts an unmeasured size.

## Explanations for the additional workbook examples

### W1 Summarize repeats

#### Workbook question · W1

Calculate the mean and **range** for the 20-row condition.

#### Teacher reasoning · W1

Teacher explanation: Mean(5+7+6)/3=6; **range** 7-5=2.

#### Teacher reasoning · W1

For the stated 20-row condition, add the run results 5, 7 and 6 to obtain 18 and divide by three to obtain mean 6. **Range** is largest minus smallest: 7 - 5 = 2.

Keep the number of **repeat**ed runs distinct from the number of data rows.

### W2 Read both curves

#### Workbook question · W2

Why can **training** error rise while validation error falls?

#### Teacher reasoning · W2

Teacher explanation: More varied **training** cases may be harder to fit exactly while giving a pattern that generalizes better.

### W3 Avoid extrapolation

#### Workbook question · W3

Can the 40-row result guarantee a mean MAE of 2.5 at 80 rows?

#### Teacher reasoning · W3

Teacher explanation: No. The observed trend does not establish a fixed proportional improvement.

### W4 Apply independently

#### Workbook question · W4

At sizes 12 and 24, validation errors are [6, 8, 7] and [4, 7, 7]. Calculate means/**range**s and a careful comparison.

#### Teacher reasoning · W4

Teacher explanation: Means 7 and 6; **range**s 2 and 3. The larger condition improves the mean by 1 but has greater observed spread in these three **repeat**s.

### W5 Explain the spread

#### Workbook question · W5

Why is an observed **range** not a confidence interval?

### Teacher reasoning · W5

Teacher explanation: It only describes the smallest and largest observed **repeat**s; no coverage-probability calculation or independent-sampling argument was supplied.

## Assess independent transfer

### Independent question · withhold the key

Selected case: At sizes 12 and 24, validation errors are [6, 8, 7] and [4, 7, 7]. Calculate means/**range**s and a careful comparison.

## Selected case answer

### Teacher key · Selected case answer

Teacher explanation: Means 7 and 6; **range**s 2 and 3. The larger condition improves the mean by 1 but has greater observed spread in these three **repeat**s.

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect and extend learning

> Connect Learning Curves to Responsible Plant Classifier, adapt support and plan a later changed-case check.

### Responsible Plant Classifier

Plot Plant Classifier train/validation performance at several **training** sizes and document the chosen next action.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

### Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

### Bridge to the next week

Next: Week 29: Feature Engineering. Retrieve today's **evidence** and identify what carries forward and what is new.

### Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **learning curve** and **repeat** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Means 7 and 6; **range**s 2 and 3. The larger condition improves the mean by 1 but has greater observed spread in these three **repeat**s. Fresh retry errors [3, 5, 7]: mean 5, **range** 4.

Require the description to say three observed runs rather than a guaranteed future interval. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Feature engineering. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

### Plan the delayed return

Suggested return points in this level: ai-4/29, ai-4/31, ai-4/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
|---|---|
| Learner code and date | |
| Concept and case used; prior exposure | |

| Record | Teacher evidence |
| --- | --- |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Learning Curves lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Read **training**-size **learning curve**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Why can **training** error rise while validation error falls?

### Supported practice answer

Reasoned practice answer: Teacher explanation: More varied **training** cases may be harder to fit exactly while giving a pattern that generalizes better.

Check every learner before discussion: Use the worked case for **Learning curve**s and **variability**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Mean(5+7+6)/3=6; **range** 7-5=2.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: A30-plant experiment gives errors [4, 4, 4]. Require a **prediction** and a reason before comparing. Expected result: Mean remains4; this additional size does not improve the observed mean over20 plants.

Reason: **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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TEACHER GUIDE · 8 OF 9

## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

> Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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## Visual concept journey

> Show one Learning Curves graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Learning Curves: the case

AI ACADEMY / AI-4/28 · 01 · LOOK AT THE FACTS

### Learning Curves: the case

**The situation**

Read a learning curve with repeated-run spread rather than drawing a certainty arrow.

**Supplied evidence**

At 10 training plants validation errors across3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set. All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

**By the end, I can explain learning curves using a traced example and a stated limitation.**

Follow the labelled picture with the matching explanation. Open the vector diagram

Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

Supplied facts: At 10 **training** plants validation errors across3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set.

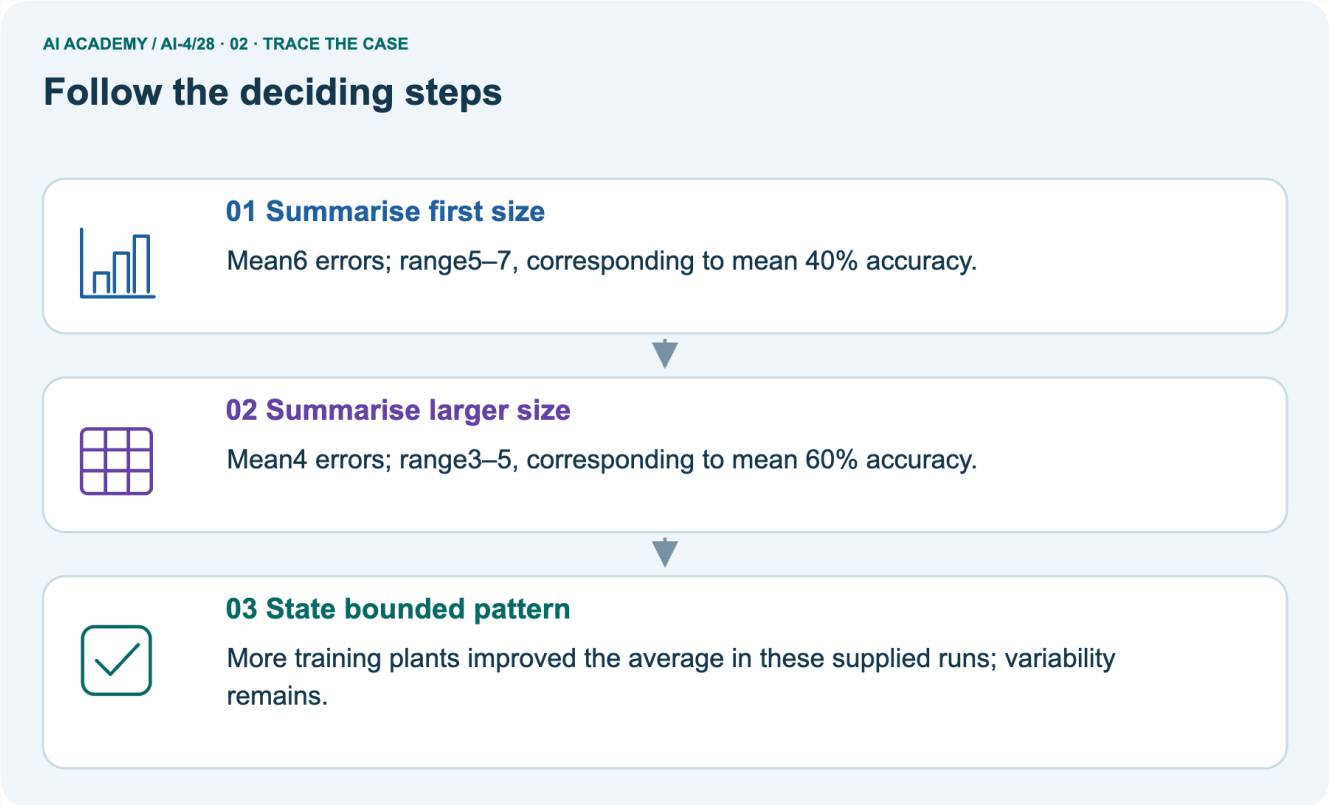
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

Summarise first size: Mean6 errors; **range**5–7, corresponding to mean 40% accuracy. → Summarise larger size: Mean4 errors; **range**3–5, corresponding to mean **60%** accuracy. → State bounded pattern: More **training** plants improved the average in these supplied runs; **variability** remains.

Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. Learning curves and variability at a glance

AI ACADEMY / AI-4/28 · 03 · INSPECT THE LESSON MODEL

### Learning curves and variability at a glance

| Part | Evidence or action | Interpretation |
| --- | --- | --- |
| 10 training rows | Validation 7 9 8 | Mean 8 range 2 |
| 20 training rows | Validation 5 7 6 | Mean 6 range 2 |
| 40 training rows | Validation 4 6 5 | Mean 5 range 2 |

The decreasing gap is encouraging in this experiment. It does not establish that doubling again will halve error. A range from three repeats is a descriptive spread, not a confidence interval or a guarantee that future results fall inside it. Repeated runs can share records and therefore are not independent studies.

Follow the labelled picture with the matching explanation. Open the vector diagram

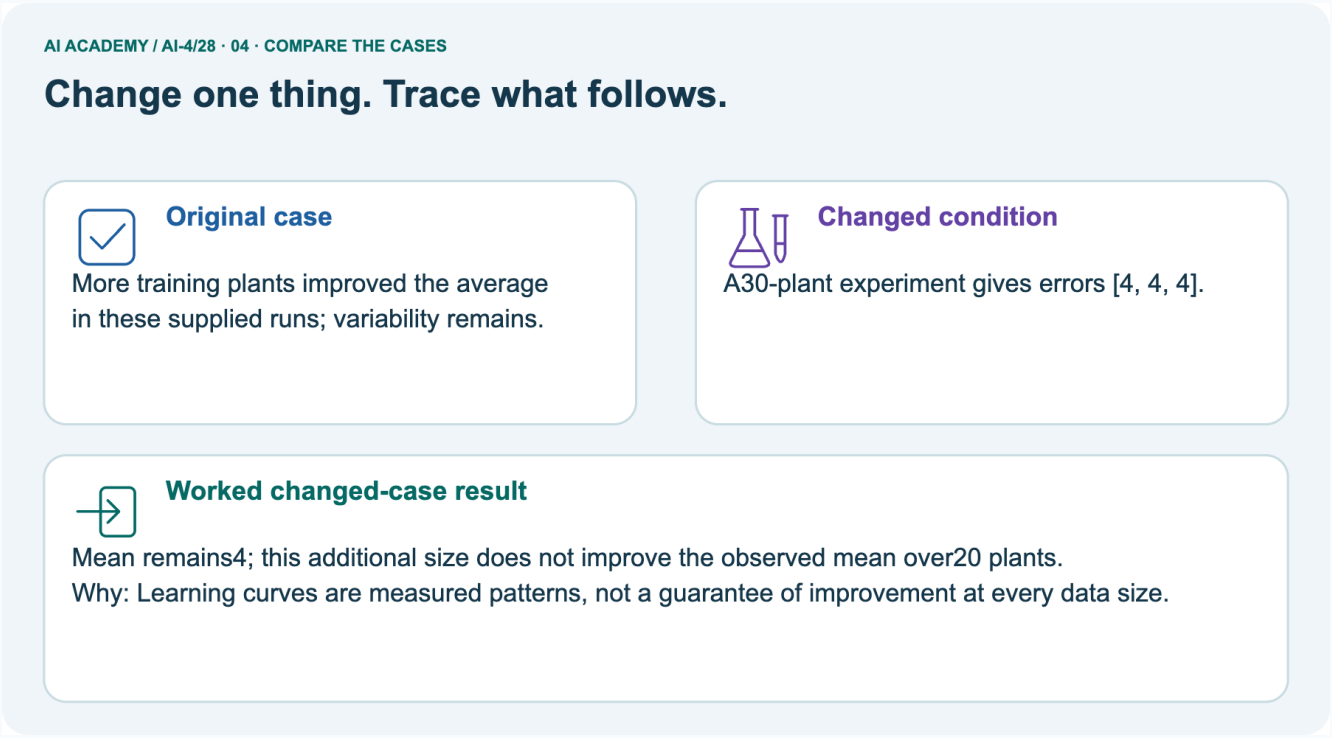
**Learning curve**s and **variability** at a glance. 10 **training** rows | Validation 7 9 8 | Mean 8 **range** 2. 20 **training** rows | Validation 5 7 6 | Mean 6 **range** 2. 40 **training** rows | Validation 4 6 5 | Mean 5 **range** 2

Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: More **training** plants improved the average in these supplied runs; **variability** remains. Changed condition: A30-plant experiment gives errors [4, 4, 4].

Expected result: Mean remains4; this additional size does not improve the observed mean over20 plants. Reason: **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

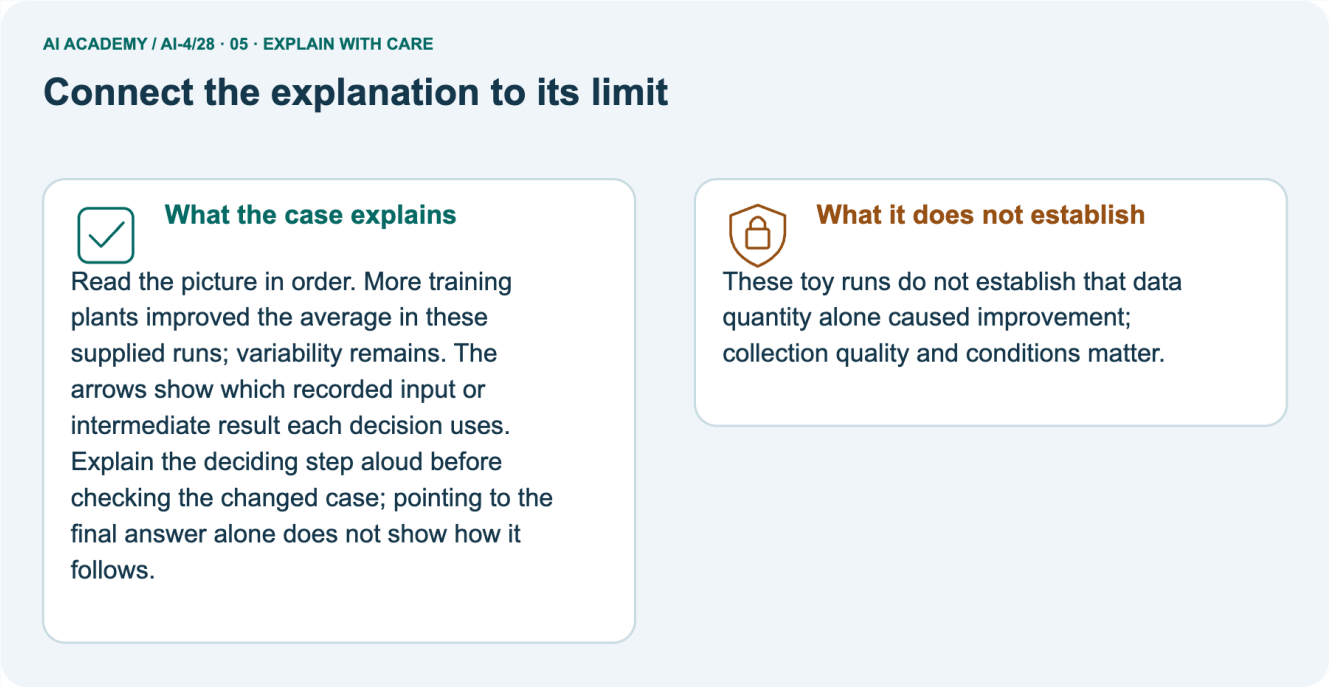
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. More **training** plants improved the average in these supplied runs; **variability** remains. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: These toy runs do not establish that data quantity alone caused improvement; collection quality and conditions matter.

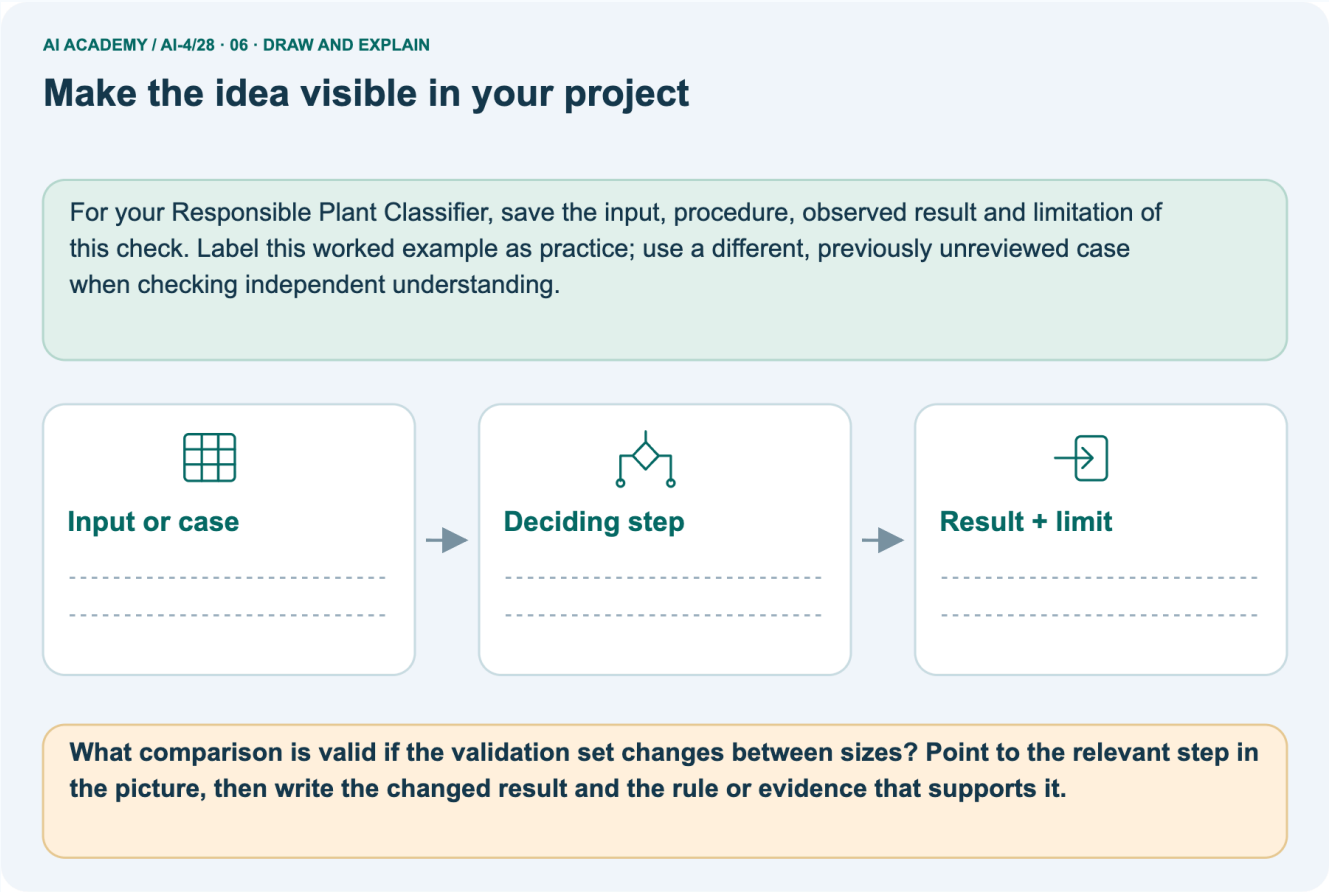
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These toy runs do not establish that data quantity alone caused improvement; collection quality and conditions matter.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The raw change is confounded by changed evaluation cases. Prefer a fixed appropriately reserved validation set and record **repeat**ed-run variation.

Changed-case check: Mean remains4; this additional size does not improve the observed mean over20 plants. **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.